checksub.

SPEAKER DIARIZATION / 29 OCT 2026

Speaker detection, measured.

pyannote Precision-3 and NVIDIA Nemotron 3 Diarization, tested on three English animation clips against a shared, human-labeled reference.

OVERALL SPEAKER ERROR

LOWER IS BETTER ↓

pyannote Precision-3

47.69%

NVIDIA Nemotron

52.30%

Precision-3 leads by 4.61 points

All three clips, including an 11-speaker stress test.
Within NVIDIA's eight-speaker limit, the scores nearly tie.

THE TEST SET				
Same audio, both detectors	03	12m 26s	5–11	2 / 3
	English animation clips	Total audio evaluated	Reference speakers per clip	Clips favor Precision-3

01 THE RESULTS

A lead overall. A different result on Veronica.

Precision-3 produces fewer total diarization errors in this pilot. NVIDIA is slightly ahead on Veronica, so the result depends on the recording.

Error rate by recording

Each point is a model's DER for one complete clip.



0-second collar · overlapping speech included

DER is an error rate, not a percentage accuracy score.

WITHIN THE EIGHT-SPEAKER LIMIT

The other two clips nearly tie.

Excluding Dave's 11-speaker stress test, the weighted difference is just **0.03 percentage points**. The overall lead is driven by the recording that exceeds NVIDIA's speaker capacity.

Precision-3

55.91%

Pizza + Veronica · weighted DER

NVIDIA

55.88%

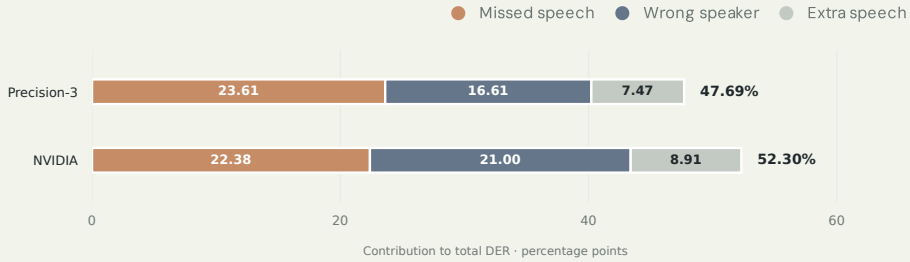
Pizza + Veronica · weighted DER

LOOKING INSIDE THE SCORE

What went wrong?

The total combines missed speech, incorrect speaker assignments, and speech detected outside the reference.

How we calculate DER ↗



02 THE RECORDINGS

Every recording tells a different story.

All three clips contain fewer detected speaker labels than the human reference. A label count alone does not establish whether a voice was assigned correctly.

01

4:23 · 11 reference voices

Dave the Barbarian

Singing, music and a large cast

Precision-3	36.06%
NVIDIA	47.24%
Speaker labels Reference / Precision-3 / NVIDIA	11 / 9 / 5

Precision-3 lower error

02

4:04 · 5 reference voices

Pizza Delivery

Rapid dialogue and background music

Precision-3	43.80%
NVIDIA	45.11%
Speaker labels Reference / Precision-3 / NVIDIA	5 / 4 / 3

Precision-3 lower error

03

3:59 · 8 reference voices

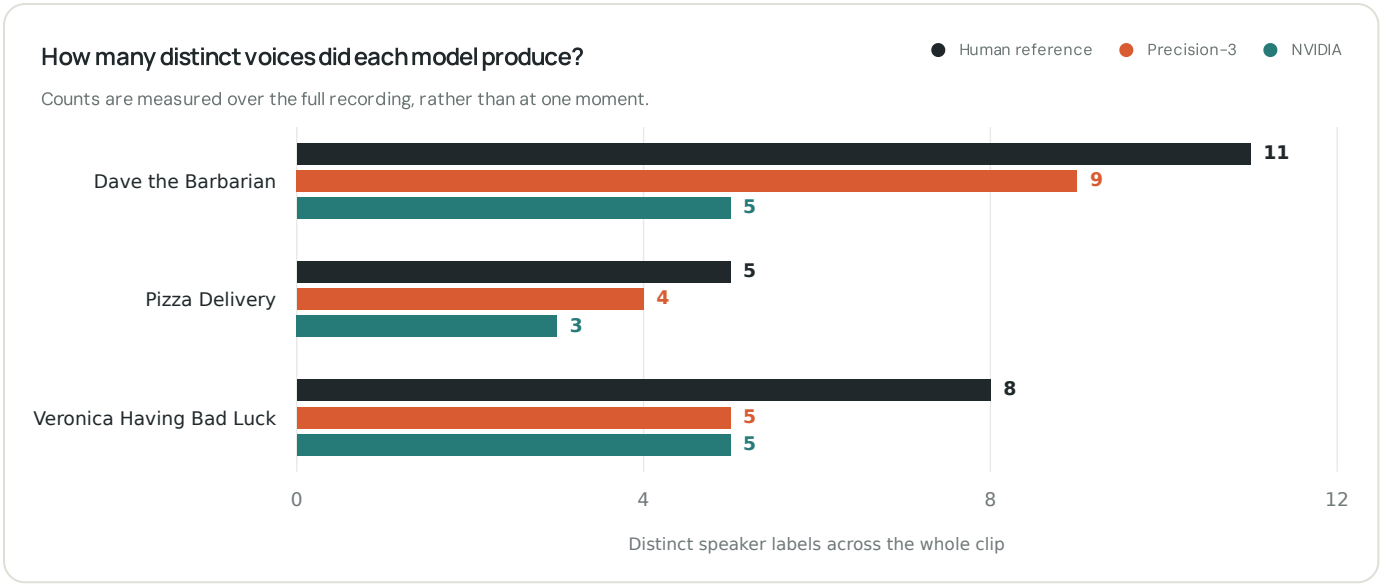
Veronica Having Bad Luck

Short turns and similar character voices

Precision-3	69.08%
NVIDIA	67.60%
Speaker labels Reference / Precision-3 / NVIDIA	8 / 5 / 5

NVIDIA lower error

ⓘ **A material limit on this comparison:** Dave has 11 reference speakers. NVIDIA's model supports at most eight speakers per session. That clip exceeds its capacity, so this is a product-fit stress test, not a fully matched model comparison.


How many distinct voices did each model produce?
Counts are measured over the full recording, rather than at one moment.
Human reference
Precision-3
NVIDIA
Dave the Barbarian
11
9
5
Pizza Delivery
5
4
3
Veronica Having Bad Luck
8
5
5
0
4
8
12
Distinct speaker labels across the whole clip

03 AN INTERACTIVE CLOSER LOOK

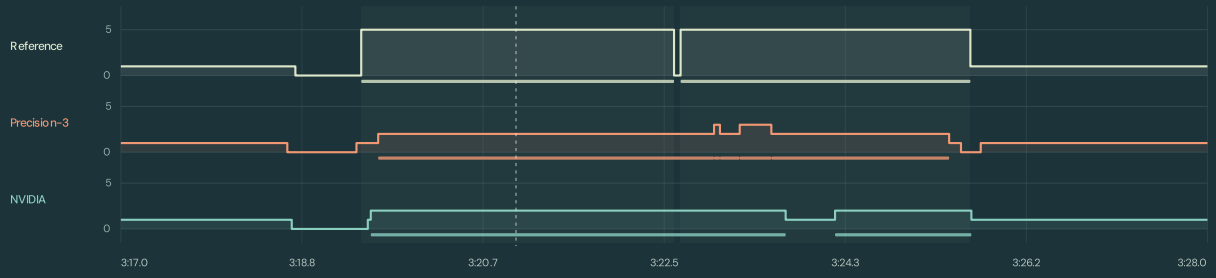
Inside the overlap.

Both models flag simultaneous voices. Explore exactly when they do, and how many speaker labels are active at the same moment.

DAVE THE BARBARIAN / FIVE-VOICE CHORUS

Five voices in the reference. Fewer in the predictions.

Human reference · Precision-3 · NVIDIA · Height = simultaneous speaker labels



3:21.00 Reference 5 Precision-3 2 NVIDIA 2

Annotated overlap

6.10s

Two chorus passages, with five reference characters singing together.

Precision-3 · within the chorus

5.71s

93.7% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

NVIDIA · within the chorus

5.50s

90.2% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

When was overlap detected?

3 intervals

TIME WINDOW	REFERENCE PEAK	PRECISION-3 PEAK	NVIDIA PEAK	REFERENCE STATUS
1:08.96–1:08.97 ↗	1	1	2	No overlap annotation
3:19.43–3:22.60 ↗	5	2	2	Annotated overlap
3:22.67–3:25.60 ↗	5	3	2	Annotated overlap

The two annotated intervals are 3:19.43–3:22.60 and 3:22.67–3:25.60. Candy, Mrs. Bogmelon, Oswidge, Faffy and Fang each have a reference row. Model-only flags outside those intervals are not independently verified. Event rows are clickable and show peak simultaneous counts inside each interval.

Detecting overlap and recovering every voice are separate tasks. In the annotated chorus, Precision-3 reaches three simultaneous labels; NVIDIA reaches two. The reference contains five. Those counts do not verify which characters were assigned correctly.

04 HOW TO READ THESE NUMBERS

These are our measured results, not the vendors' published benchmark scores. They describe this test set

A useful pilot. A specific set of limits.

and these references.

THE METRIC

Diarization error rate

$$\frac{\text{Missed speech + extra speech + wrong-speaker time}}{\text{Reference speaker time}}$$

Multiplied by 100. **Lower is better.** DER can exceed 100%; subtracting it from 100 does not give a reliable "accuracy" score.

Scores are weighted by reference speaker time. Overlapping reference voices each contribute speaker time.

01 Identical audio, isolated detection

Both models received the same 16 kHz mono WAV for each clip. This compares Precision-3 with Nemotron 3 Diarization, not the full Nemotron 3.5 Transcribe pipeline.

02 Human names, subtitle-based times

Speaker identities were filled by a human in Checksub subtitle exports. Speech boundaries were not independently annotated. Display padding and missing intervals can inflate timing errors. Only the Dave chorus has explicit overlap reference labels.

03 Strict, consistent scoring

Zero boundary collar; overlap included; full-clip evaluation. Speaker IDs are matched to names globally and one-to-one, so splitting or merging identities is penalized.

04 A small, specialized sample

Three English animation clips, with music and exaggerated voices. Dave exceeds NVIDIA's eight-speaker capacity. These results do not establish performance on customer recordings, other languages, or recordings with matched speaker budgets.

05 Speed and cost need a separate test

Precision-3 used a hosted API; NVIDIA used local GPU inference. Queueing, upload and model load have different scopes. This report makes no apples-to-apples speed or cost claim.

05 CONCLUSION

Keep Precision-3 as the baseline, for now.

There is no broad quality winner established yet. Precision-3 leads across all three clips, but **the two recordings within NVIDIA's speaker limit are nearly tied**.

Retain the current baseline while running a larger, independently timed comparison. NVIDIA's local deployment and competitive results within its capacity justify that next test. Neither system recovers all five chorus voices here.

● PROVISIONAL RECOMMENDATION

BEFORE A REPLACEMENT DECISION

1. **Tighten the reference**
   Annotate actual speech boundaries and all overlaps, independently of subtitle display timing.
2. **Broaden and match the test set**
   Add customer-style interviews in our working languages; compare recordings within the eight-speaker limit, with larger casts assessed separately.
3. **Measure the complete trade-off**
   Compare correction time, deployment cost and latency under consistent conditions, then inspect the final Checksub editor output.

checksub.